

Marishka Manoj Mehta

PhD Candidate · Computational Psychiatry

Laureate Institute for Brain Research · Tulsa, OK · marishka.mehta@gmail.com

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Education

PhD, Cyber Studies (Specialization: Data Science and Psychology) Aug 2022 – Dec 2026 (expected)

The University of Tulsa

M.A., Psychology 2019 – 2021

New York University, Graduate School of Arts and Sciences

Thesis: Examining neural correlates of reward and punishment learning in depression · advised by Dr. Laurel Morris (Morris Lab, Mount Sinai).

Post Graduate Diploma, Clinical and Community Psychology 2018 – 2019

The Maharaja Sayajirao University of Baroda, India

M.Sc., Neuroscience (Neurasmus Dual Diploma) 2015 – 2019

University of Bordeaux (France) & Laval University (Canada)

- Bordeaux (2015–2017) · *Thesis:* Impact of stress and its interaction with high-fat diet on memory in corticosteroid-binding globulin (CBG) deficient mice.
- Laval (2017–2019) · *Thesis:* Role of Slitrk2 and Slitrk5 in axonal growth and guidance in midbrain dopaminergic neurons.

B.Sc. (Hons.), Biomedical Science 2012 – 2015

Queen Mary, University of London

Fellowships, Awards & Honors

Dissertation Completion Fellowship Jun 2026 – May 2027

The University of Tulsa

Trainee Professional Development Award (\$1,000) Nov 2025

Society for Neuroscience

Distinguished Graduate Research Award — nominee Apr 2025

The University of Tulsa

Waived Membership, Early Career Scientists Dec 2024

Society of Biological Psychiatry

Hurricane Pitch Competition — 2nd Place (\$3,500) 2024

The University of Tulsa

Honorable Mention

May 2021

NYU Psychology Masters Research Conference

FidEX Scholarship (€9,000)

Oct 2016 – Jul 2017

University of Bordeaux

Science and Engineering Excellence Scholarship (£15,000)

Sep 2012 – May 2015

Queen Mary, University of London

Research Positions

Graduate Research Assistant

Aug 2022 – present

Laureate Institute for Brain Research, Tulsa, OK · PI: Dr. Ryan Smith

Clinical Research Coordinator

Jul 2021 – Jul 2022

Depression and Anxiety Center, Icahn School of Medicine at Mount Sinai, New York, NY · PI: Dr. Laurel Morris

Graduate Teaching Assistant, Scientific Programming

Jan–May 2021, Jan–May 2022

New York University, New York, NY

Peer-Reviewed Publications & Book Chapters

Author name in **bold**. Numbered in reverse chronological order.

2025

1. **Mehta, M. M.**, Hakimi, N. H., Pena, O., Torres, T., Goldman, C. G., Lavalley, C., Stewart, J. S., Berg, H., Ironside, M., Paulus, M. P., Aupperle, R., & Smith, R. (2025). Computational mechanisms of approach-avoidance conflict predictively differentiate between affective and substance use disorders. *Computational Psychiatry*. <https://doi.org/10.5334/cpsy.131>
2. **Mehta, M. M.**, Paulus, M. P., & Smith, R. (2025). Computational approaches for uncovering interoceptive mechanisms in psychiatric disorders and their biological basis. In *Perceptual Dysregulation in Psychiatric Nosology*. https://doi.org/10.1007/7854_2024_572
3. Lavalley, C. A., **Mehta, M. M.**, Taylor, S., Chuning, A. E., Stewart, J. L., Huys, Q. J. M., Khalsa, S. S., Paulus, M. P., & Smith, R. (2025). Computational mechanisms underlying multi-step planning deficits in methamphetamine use disorder. *Translational Psychiatry*. <https://doi.org/10.1038/s41398-025-03390-8>
4. Hodson, R., **Mehta, M.**, Taylor, S., Lavalley, C., Stewart, J. S., Guinjoan, S. M., Ironside, M., White, E., Kupulicki, R., Paulus, M. P., & Smith, R. (2025). Attenuated learning rates for negative outcomes in substance use disorders: A replication and extension of prior longitudinal computational modeling results. *Drug and Alcohol Dependence*. <https://doi.org/10.1016/j.drugalcdep.2025.112922>
5. Beltrán, J., Jacob, Y., **Mehta, M. M.**, Hossain, T., Adams, A., Fontaine, A., Torous, J., McDonough, C., Johnson, M., Delgado, A., Murrough, J. W., & Morris, L. S. (2025). Digital measures of activity and motivation impact depression and anxiety in the real world. *npj Digital Medicine*. Link

6. Goldman, C. G., Hakimi, N., **Mehta, M. M.**, Ironside, M., Paulus, M. P., & Smith, R. (2025). Pavlovian bias is associated with symptom severity in individuals with anxious depression. *Translational Psychiatry*. Link

2024

7. **Mehta, M. M.**, Butler, G., Ahn, C., Whitaker, Y., Bachi, K., Jacob, Y., Treadway, M., Murrough, J. W., & Morris, L. S. (2024). Intrinsic and extrinsic control impacts motivation and outcome sensitivity: The role of anhedonia, stress, and anxiety. *Psychological Medicine*.
<https://doi.org/10.1017/S0033291724002022>

2023

8. Hodson, R., **Mehta, M. M.**, & Smith, R. (2023). The empirical status of predictive coding and active inference. *Neuroscience and Biobehavioral Reviews*.
<https://doi.org/10.1016/j.neubiorev.2023.105473>
9. **Mehta, M. M.**, Na, S., Gu, X., Murrough, J. W., & Morris, L. S. (2023). Reward-driven agency increases with age but is reduced in patients with depression and anxiety disorders. *PLOS ONE*, 18(3), e0282727. <https://doi.org/10.1371/journal.pone.0282727>

2022

10. Morris, L. S., **Mehta, M. M.**, Ahn, C., Corniquel, M., Verma, G., Delman, B., Hof, P. R., Jacob, Y., Balchandani, P., & Murrough, J. (2022). Ventral tegmental area integrity measured with high-resolution 7-Tesla MRI relates to motivation across diagnoses. *NeuroImage*, 264, 119704.
<https://doi.org/10.1016/j.neuroimage.2022.119704>
11. Morris, L. S., Grehl Mora, M., Rutter, S., **Mehta, M. M.**, & Westwater, M. L. (2022). On what motivates us: A detailed review of intrinsic versus extrinsic motivation. *Psychological Medicine*, 1–16. <https://doi.org/10.1017/S0033291722001611>

2019

12. Medeiros, G. F., Lafenêtre, P., Janthakhin, Y., Cerpa, J. C., Zhang, C., **Mehta, M. M.**, Mortessagne, P., Helbling, J. C., Ferreira, G., & Moisan, M. P. (2019). Corticosteroid-binding globulin deficiency specifically impairs contextual and recognition memory consolidation in male mice. *Neuroendocrinology*, 109(4), 322–332. <https://doi.org/10.1159/000499827>

Workshop Papers

- Mehta, M. M.**, & Dennis, D. K. (2026). When Old Rules Override New Decisions: Policy-State Revision in LLM Checkers. NeurIPS 2026 Workshop on Interpreting Agent Behavior. Accepted.

Preprints

- Beltrán, J., Li, M., Jacob, Y., Butler, G., **Mehta, M.**, Murrough, J. W., Radulescu, A., & Morris, L. S. (2025). VTA network dominance in depression confers distinct psychopathological states through blunted neural tracking of reward prediction errors. *Research Square*.
<https://doi.org/10.21203/rs.3.rs-7166201/v1>

Manuscripts in Preparation

- **Mehta, M. M.**, Lavalley, C. A., Chuning, A., Stewart, J. L., Huys, Q. J. M., Khalsa, S. S., Paulus, M. P., & Smith, R. Anxious Interoceptive Arousal Modulates Aversive Pruning Through Directed Insula-To-Prefrontal Coupling Across Healthy and Anxious-Depressed Individuals. *In preparation*.
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Invited Talks

1. **Neurocomputational mechanisms of maladaptive avoidance.** Graduate Research Celebration Day, The University of Tulsa · April 14, 2025 · *Distinguished Graduate Research Award nominee*.
 2. **Pre-registered replication of longitudinal computational modeling results confirming differences in approach-avoidance conflict behavior within anxiety, depression, and substance use disorders.** Computational Psychiatry Conference · Minneapolis, MN · 2024.
 3. **Comparing denoising approaches in ultra-high field resting-state fMRI.** Center for Computational Psychiatry · 2021.
 4. **Role of Slitrk2 and Slitrk5 on axonal growth and guidance in midbrain dopaminergic neurons.** 6th Annual Neurasmus Meeting · Göttingen, Germany · 2017.
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Selected Posters

1. **Mehta, M. M.**, Lavalley, C. A., Chuning, A., Stewart, J. L., Huys, Q. J. M., Khalsa, S. S., Paulus, M. P., & Smith, R. (Nov 2025). *Neurocomputational mechanisms of cognitive avoidance during prospective planning: effects of interoceptive anxiety induction in affective disorders*. Society for Neuroscience, San Diego, CA.
2. **Mehta, M. M.**, Sanchez, O., Hakimi, N., Torres, T., Lavalley, C. A., Taylor, S., Stewart, J. L., Khalsa, S. S., Berg, H., Ironside, M., Paulus, M. P., Aupperle, A., & Smith, R. (Jul 2024). *Pre-registered replication of longitudinal computational modelling results confirming differences in approach-avoidance conflict behavior within anxiety, depression, and substance use disorders*. Computational Psychiatry Conference, Minneapolis, MN.
3. **Mehta, M.**, Jacob, Y., & Morris, L. (May 2022). *Comparing denoising approaches in ultra-high field resting-state fMRI*. Society of Biological Psychiatry, New Orleans, LA.
4. **Mehta, M.**, Tubridy, S., & Morris, L. (May 2021). *Neural correlates of reward and punishment learning in major depressive disorder*. 25th Annual NYU Psychology Masters Research Conference, New York, NY. *Honorable Mention*.
5. **Mehta, M.**, Salesse, C., Charest, J., Doucet-Beaupré, H., De Koninck, P., & Lévesque, M. (Jul 2017). *Role of Slitrk2 and Slitrk5 on axonal growth and guidance in midbrain dopaminergic neurons*. 6th Annual Neurasmus Meeting, Göttingen, Germany.

Teaching & Mentoring

Teaching

- Graduate Teaching Assistant, Scientific Programming — New York University · Jan–May 2021, Jan–May 2022
- Python workshops for Master's students — Morris Lab, Mount Sinai · 2021–2022

Mentoring

- Graduate and undergraduate students — Laureate Institute for Brain Research · 2022–present
 - High school student — Morris Lab, Mount Sinai · 2021–2022
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Professional Service

Peer review

- Ad hoc reviewer: *Psychological Medicine* · *Nature Mental Health* · *Scientific Reports* · *Communications Biology*

Judging & panels

- Poster judge, Research Day, Laureate Institute for Brain Research · 2025
 - Panelist, *Peer Review and Publishing Panel* — Student Fellowship and Training Subcommittee, Laureate Institute for Brain Research · 2025
 - Volunteer, Scientific Review Committee — Tulsa Regional Science Fair, Tulsa Regional STEM Alliance · 2024
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Technical Skills

Cognitive modeling · Model-based/model-free reinforcement learning, active inference, DCM/PEB/BMR

Neuroimaging · 3T/7T fMRI; fMRIPrep, FSL, SPM, AFNI, MEICA, Nilearn

LLM evaluation · Behavioral evaluation, LLM-as-a-judge assessment, prompt ablations, evaluation task design, bias and reliability analysis, sampling-parameter sweeps; Qwen and Llama

Statistics & machine learning · Bayesian hierarchical modeling (VBA), mixed-effects models, path analysis, ensemble classification (caretEnsemble), cross-validation, sensitivity analysis; SEM/EFA/ICC

Programming & data analysis · Python (NumPy, pandas, SciPy, Matplotlib, Jupyter), R, MATLAB

Research pipelines & reproducibility · Automated scoring, dataset validation, experiment tracking, data provenance, resumable collection, Git, mock-based testing, preregistration; Python automation

Model inference & research computing · vLLM, Ollama, hosted model APIs, SLURM-based HPC

Experimental methods · PsychoPy, jsPsych, REDCap

Clinical research · Structured interviews, symptom assessments, IRB coordination

References

Available upon request.